INCH-POUND

MIL-I-48632 (AR)
AMENDMENT 2
05 May 1993
SUPERSEDING
AMENDMENT 1
23 July 1988

MILITARY SPECIFICATION

INTEGRATED CIRCUIT, DIGITAL, CMOS CONTROL AND TIMING BASE, MONOLITHIC, SILICON

This Amendment forms a part of Military Specification MIL-I-48632 (AR), dated 31 July 1986, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1: Change: "M718E1/M741E1" to "M718A1/M741A1".

PAGE 2

* 2.1.2, Third line down: Delete "extent" and substitute "extend".

PAGE 3

* 3.2.1: Delete in its entirety and substitute:

"3.2.1 Dimensions. The package dimension shall conform to MIL-M-38510, Appendix C, Type Designation D-2, Configuration 1."

3.2.2:

Second line down: Delete "18 pin" and substitute "16 pin".

* Third line down: Delete "figure 16" and substitute "figure 1".

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:35Z
Check the source to verify that this is the current version before use.

* Add new paragraph:

"3.2.6 Preseal bake. Eutectic attach substrate and sealing glass shall be baked at 125°C in a dry nitrogen atmosphere for a specified interval (16 hours minimum for eutectic attached substrate and 2 hours minimum for sealing glass) immediately prior to sealing."

PAGE 4

* 3.4.1: Delete in its entirety and substitute:

"3.4.1 Maximum rating. (As specified in Table I)

TABLE I MAXIMUM RATING

Characteristic	Symbol	Rating	Units
DC Supply Voltage (See 3.4.4)	V_B	5 to +10	Volts
All inputs	V_{IN}	-0.5 to +10	Volts
Temperature			
Operating	$T_A(OP)$	-55 to +125	°C
Storage	$T_A(STG)$	-65 to +150	°C

3.4.2:

Delete "$-32°C \leq T_A \leq +63°D$ $V_B=7.5v$" under conditions and substitute "$-32°C \leq T_A \leq +63°C$ $V_B=7.5v$".

* Delete "Table 2" and substitute "Table II".

* Delete "TABLE 2" and substitute "TABLE II".

* Last line: Delete "See Figure 7" and substitute "See 4.5.4.1".

* Delete "Capacitance coupled feedback signal amplitude Terminal 10" characteristic in its entirety.

Source: https://assist.dla.mil -- Downloaded: 20260-22T19:35Z
Check the source to verify that this is the current version before use.

3.4.2:

Under conditions: Delete "(Greater of LVD or 2.5 volts - V_B - 7.5 volts)" and substitute "(Greater of LVD or 2.5 volts $\leq$ T_A $\leq$ 7.5 volts)"

Delete "-32° - T_A - +63°C" and substitute "-32° $\leq$ T_A $\leq$ +63°C".

Delete "-32°C - T_A - +63°C" and substitute "-32°C $\leq$ T_A $\leq$ +63°C".

3.4.2:

* Add new characteristic:

"Characteristic	Conditions	Min.	Max.	Unit
Leakage Current (Pin 2)	T_A=+25°C V_B=7.5V		15.0	UA
	See 4.5.2.1 -32°C-T_A- + 63°C V_B=7.5		15.0	UA

* "0" Output Voltage: Terminal 15: Delete "Figure 8" and substitute "See 4.5.4.2".

* Sequential Operation:

Delete "Greater of LVD or 2.5" and substitute "Greater of LVD + 0.5V or 2.5".

Delete "-32° $\leq$ T_A $\leq$ +63°C" and substitute "32°C $\leq$ T_A $\leq$ 63°C".

Add "See 4.5.5" under conditions.

Delete "(Greater of LVD or 2.5 volts $\leq$ T_A $\leq$ 7.5 volts)" and substitute "(Greater of LVD or 2.5 volts $\leq$ V_B $\leq$ 7.5 volts)".

3.4.2:

* LVD Function Voltage:

Delete "LVD Function Voltage" and substitute "LVD Function Voltage: Terminal 15".

Delete "-32°C T_A + 63°C" and substitute "-32°C $\leq$ T_A $\leq$ 63°C".

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:35Z
Check the source to verify that this is the currentversion before use.

* Delete ""0" Output Voltage:" and substitute ""0" Output Voltage: Terminal 11".

* Input Switching Voltage: Terminal 10: Delete in its entirety and substitute:

"Characteristic	Conditions	Min	Max	Units
Input Switching Voltage: Terminal 10	See 4.5.7 and 3.4.8	3.3	6.3	V
Difference between positive going threshold and negative going threshold: Terminal 10"		1.0	-	V

* Input Resistance: Delete in its entirety and substitute:

"Characteristic	Conditions	Min	Max	Units
Input Resistance	Terminal 1, 7 Connected to V_B thru 500K Ohms	2.0	6.9	V
	Terminal 5 Connected to V_B thru 500K Ohms	4.0	6.9	V
	Terminal 2 Connected to Ground thru 500K Ohms See 4.5.8"	1.5	3.75	V

* Add new paragraph:

"3.4.2.2 Capacitance coupled feedback test characteristics. The following performance characteristics apply; while functioned in accordance with paragraph 3.4.2.1.

Characteristic	Conditions	Min	Max	Units
Capacitance coupled feedback signal amplitude: Terminal 10	V_B=5.0V with terminal driven by a simulated TTB signal source (see 4.5.12)	0.4	-	V

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:35Z
Check the source to verify that this is the current version before use.

3.4.2.2.1 Sampling Plan for capacitance coupled feedback test. The I.C. shall be sampled in accordance with MIL-STD-105, Inspection Level II with an AQL of 0.04% and an acceptance/rejection criteria of 0-1."

* 3.4.3, last line: Delete "This test to be...rate" and substitute "This test pattern to be...rate".

PAGE 8

* Figure 1A: Delete "415" and substitute "448".

PAGE 9

3.4.5:

* Last line: Delete "KHz Max" and substitute "KHz".
* Add: Function and testing to 50KHz is acceptable".

* 3.4.9.1, Third line down: Delete "26" and substitute "16".

PAGE 10

* Figure 2: Delete "Para 3.3.6" and substitute "Para 3.4.5".

PAGE 12

3.4.9.2:

First paragraph: Delete "Para. 3.3.9.3" and substitute "Para. 3.4.9.3".

Paragraph designated by *: Delete "3.3.9.4" and substitute "3.4.9.4" in 2 places.

3.4.9.2:

* Third paragraph: Delete "Count clk...1024. Count PTB...1024." and substitute "Count CLK periods (T) during PTB pulse duration (64T) for count $\leq$ 1024. Count PTB periods (38T) for count $>$ 1024."

* 3.4.9.3: Delete in its entirety and substitute:

"3.4.9.3 Input Terminal 4. The input voltage levels applied to the Input Terminal 4 during the sequence test must equal:

High State: V_B - 1V ($\pm$ 50MV)
Low State: 1V - ($\pm$ 50 MV)

for all values of supply voltage."

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:35Z
Check the source to verify that this is the current version before use.

PAGE 13

* 3.5.1: Delete "paragraph 3.5.15" and substitute "paragraph 3.4.2".

* 3.5.2: Delete "paragraph 3.5.15" and substitute "pargraph 3.4.2".

* 3.5.5: Delete "paragraph 3.5.15" and substitute "paragraph 3.4.2".

* 3.5.6:

Delete "paragraph 3.5.15" and substitute "paragraph 3.4.2".

Delete "Method 2002" and substitute "Method 2001".

PAGE 14

* 3.5.15: Delete in its entirety.

* 3.5.16: Delete in its entirety.

* 3.5.19: Delete in its entirety.

PAGE 15

TABLE III:

* Process Screening: Delete "3.5.18" and substitute "3.5.1, 3.5.2, 3.5.4, 3.5.6, 3.5.13, 3.5.17"" under REQ PARA.

* Solderability: Delete "1010" and substitute "2003" under METHOD PARA.

* Leakage current: Add "4.5.2.1 15.0" to the METHOD PARA and the LIMITS column.

* Group II: Add "T_A = +25°C" into the EXAMINATION OR TEST column.

REQ PARA: Delete "3.5.15" and substitute "3.4.2" in 4 places.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T19:35Z
Check the source to verify that this is the currentversion before use.

* TABLE III (continued):

Delete "Input Resistance Terminal 5 & 7" in its entirety and substitute:

*Examination or Test	Conditions	Req Para	Meth Para	Limits Min	Limits Max	Units
Input Resistance Terminal 1& 7	V_B=7.5	3.5.15	4.5.8	2.0	6.9	V

Add the following test:

*Exam.	Cond.	Req. Para.	Meth Para.	Limits Min.	Limits Max.	Units	No. of Sample To be Insp.
Input Resistance Term. 5	V_B=7.5V	3.4.2	4.5.8	4.0	6.9	V	LPTD 3
Input Resistance Term. 2	V_B=7.5V	3.4.2	4.5.8	1.5	3.75	V	LPTD 3
Capacitance Feedback Test*	V_b=5.0V	3.4.2	4.5.12	See 3.4.2.2	see 3.4.2.2.1		

*Under REQ PARA: Delete "3.5.15" and substitute "3.4.2" in 6 places.

TABLE III (Continued):

* Leakage Current: Add "4.5.2.1 15.0" to the METHOD PARA and the LIMITS column.

* "1" Output Voltage Terminal 11: Delete "-2.5" and substitute "25 -" under MIN MAX column.

"1" Output Voltage Terminal 11: Delete 4.5.4.2" and substitute "4.5.9".

"0" Output Voltage Terminal 11: Delete "4.5.9" and substitute "4.5.10".

* Under REQ PARA: Delete "3.5.15" and substitute "3.4.2" in 8 places.

* Under LIMITS: Delete "25" and substitute "2.5" in 2 places.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:35Z
Check the source to verify that this is the current version before use.

PAGE 18

* Under REQ PARA: Delete "3.5.15" and substitute "3.4.2" in 7 places.

* LVD Function Voltage: Under requirement para., delete "3.5.16" and substitute "3.4.2".

"0" Voltage Terminal 11: Delete "4.5.9" and substitute "4.5.10".

PAGE 19

4.3.2: Delete "TABLE V" and substitute "TABLE III".

4.3.3: Delete in its entirety and substitute the following:

"4.3.3 Inspection conditions. Unless otherwise specified, all examinations and tests shall be made under the following conditions:

a.	Temperature	25 ± 5 Degrees C
b.	Humidity	10 to 75% RH
c.	Pressure	29 ± 3 inches of Hg
d.	Voltage an current measure	±2%"

4.4: Delete "TABLE VI" and substitute "TABLE IV".

PAGE 20

TABLE IV:

* Solderability: Delete "1010" and substitute "2003" under METHOD PARA.

In-Process Screening: Under REQ PAPA, delete "3.5.18" and substitute "3.5.17".

* Under REQ PARA: Delete "3.5.15" and substitute "3.4.2" in 2 places.

PAGE 21

* 4.4.2: Delete "Design activity" and substitute "technical agency".

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:35Z
Check the source to verify that this is the current version before use.

* 4.5.2: Delete "Figure 5" and substitute "Figure 5A" in 2 places.

* Add new paragraph:

"4.5.2.1 Leakage Current (pin 2). The microcircuit shall be connected as shown in Figure 5B. The supply voltage shall be applied between terminals 16 (VB) and 8 (ground). A minimum of 50 milliseconds shall elapse before measuring the leakage current. The leakage current shall be as specified in Table II".

* Figure 5: Delete in its entirety and substitute new Figure 5A and 5B included.

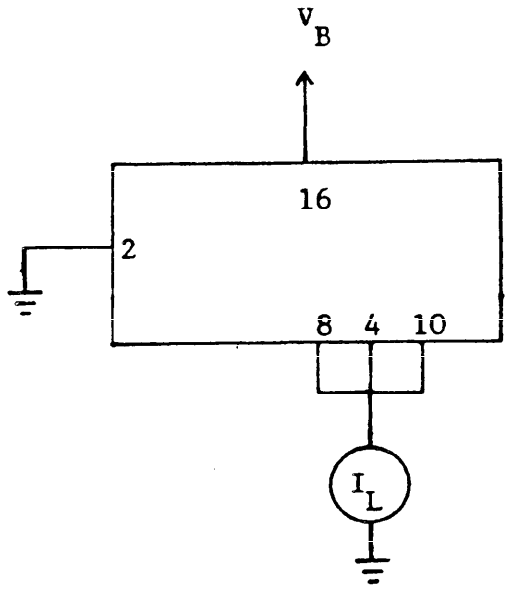


FIGURE 5A

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:35Z
Check the source to verify that this is the current version before use.

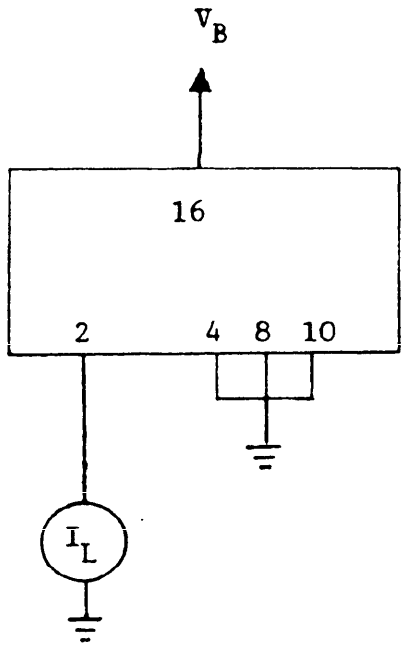


FIGURE 5B

Source: https://assist.dla.mil -- Downloaded: 20261O-22T19:35Z
Check the source to verify that this is the current version before use.

PAGE 23

* 4.5.7: Add the following at the end of paragraph:

"The voltage level on terminal 10 shall be slowly decreased. The level on pin 11 shall be observed and when pin 11 goes from a low to a high level, the voltage level on terminal 10 shall be measured as the negative going threshold on pin 10."

PAGE 26

* 4.5.11:

Second line down: Delete "125°C" and substitute "125°C minimum".

Third line down: Delete "Figure 16" and substitute "Figure 14".

PAGE 27

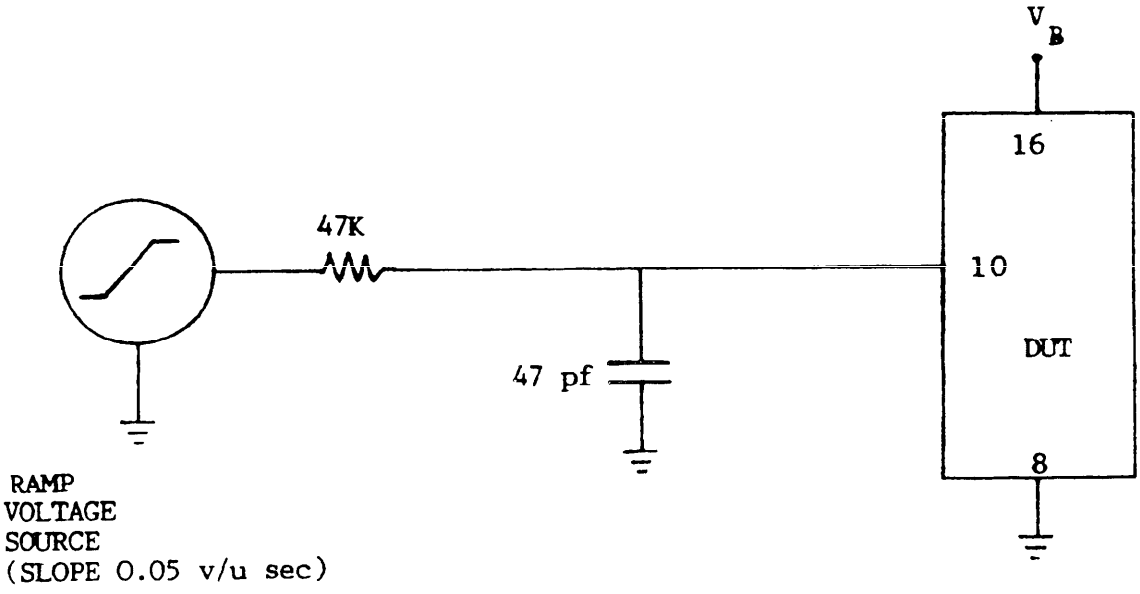
* 4.5.12:

Second line down: Delete "between the output at terminal 9 and the input".

Last line: Delete "Table 2" and substitute" paragraph 3.4.2.2".

* FIGURE 15: Delete in its entirety and new figure 15 included.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T19:35Z
Check the source to verify that this is the currentversion before use.



NOTE 1. FEED-BACK VOLTAGE

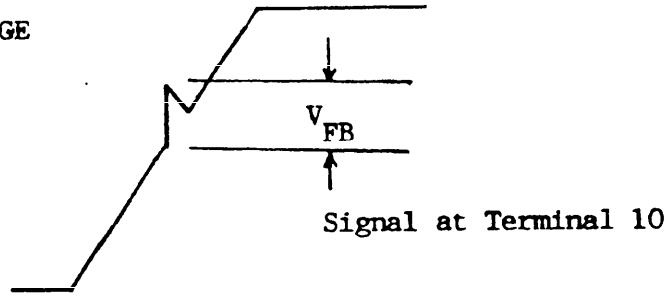


FIGURE 15. FEEDBACK SIGNAL TEST

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:35Z
Check the source to verify that this is the current version before use.

PAGE 28

* 4.6:

Second line down: Delete "method 5004" and substitute "method 5004 Class B".

Burn-In Test: Delete "125°C" and substitute "125°C minimum".

Post Burn-In: Delete "Table I" and substitute "Table II". Delete "125°C" and substitute "25°C".

PAGE 29

* 6.4: Delete "ammunition data cards" in 2 places.

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1375-A506)

Source: https://assist.dla.mil -- Downloaded: 20261O-22T19:35Z
Check the source to verify that this is the current version before use.